

Lab 6: Gradebook

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The below is my automated gradebook project for lab06:

```
# Reading in the csv file for the gradebook
data <- read.csv(file="student_homework.csv", row.names = 1)

# Other gradebook vectors for testing
student1 <- c(100, 100, 100, 100, 100, 100, 100, 90)
student2 <- c(100, NA, 90, 90, 90, 90, 97, 80)
student3 <- c(90, NA, NA, NA, NA, NA, NA, NA)
```

Question 1

The grade function:

```
grade <- function(scores)
{
  # Making a copy of the vector to modify
  scores_for_grading <- scores

  # is.na() creates a logical vector for scores in the input vector that are missing
  # We then replace those scores with -1 (lower than any possible score) to make sure that is
  scores_for_grading[is.na(scores_for_grading)] <- -1

  # Check which score is the lowest
  lowest_index <- which.min(scores_for_grading)
  scores_without_lowest <- scores_for_grading[-lowest_index]

  # If there are any unresolved NAs, we remove them
  scores_without_lowest <- scores_without_lowest[scores_without_lowest != -1]

  # Compute the mean
  mean(scores_without_lowest)
}
```

Testing:

```
student3_grade <- grade(student3)
print(student3_grade)
```

```
[1] 90
```

Question 2

To find out who has the top scoring student in the class is, we have to calculate average scores for everyone and determine who has the highest one.

```
# This applies the grade function to each row of the gradebook, i.e. once for every student,
final_grades <- apply(data, 1, grade)
print(final_grades)
```

student-1	student-2	student-3	student-4	student-5	student-6	student-7
91.75	82.50	84.25	84.25	88.25	89.00	94.00
student-8	student-9	student-10	student-11	student-12	student-13	student-14
93.75	87.75	79.00	86.00	91.75	92.25	87.75
student-15	student-16	student-17	student-18	student-19	student-20	
78.75	89.50	88.00	94.50	82.75	82.75	

```
top_scorer_index <- which.max(final_grades)
print(top_scorer_index)
```

```
student-18
18
```

Therefore, the top scoring student in the class is Student 18.

Question 3

To find out which homework was the toughest, we have to calculate average scores for each and determine which led to students performing the worst.

```
# This applies the grade function to each column of the gradebook, i.e. once for every homework
final_grades_per_hw <- apply(data, 2, grade)
print(final_grades_per_hw)
```

hw1	hw2	hw3	hw4	hw5
89.36842	80.88889	81.21053	89.63158	83.42105

```
lowest_score_index <- which.min(final_grades_per_hw)
print(lowest_score_index)
```

```
hw2
2
```

Therefore, the toughest homework was Homework 2.